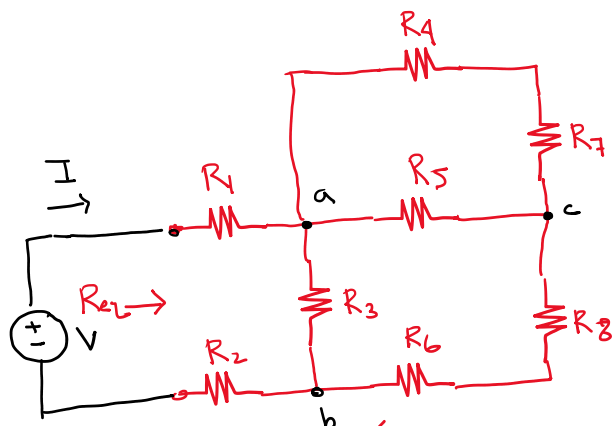
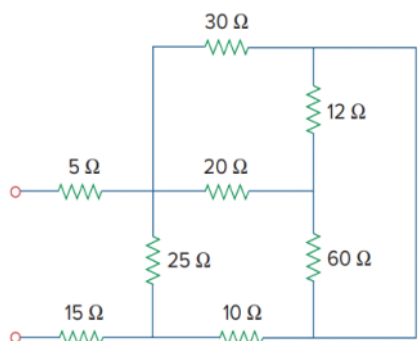


Equivalent CKT tutorial

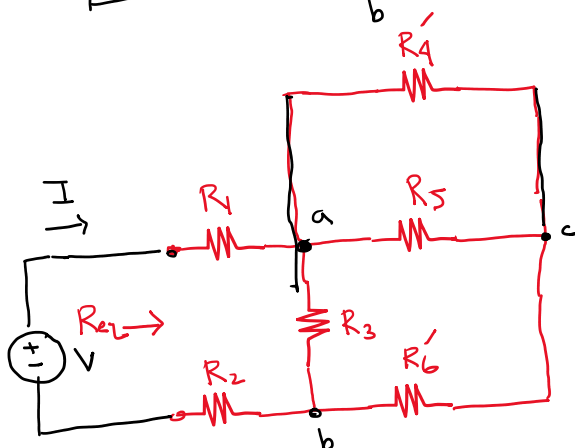
Friday, March 6, 2026 10:34 AM

point 3 or more elements are connected



$$b-c: R_6' = R_6 + R_8$$

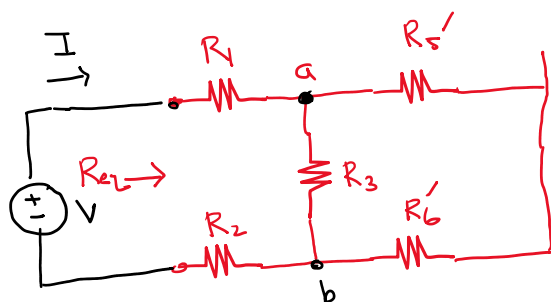
$$a-c: R_4' = R_4 + R_7$$



$$a-c: R_5$$

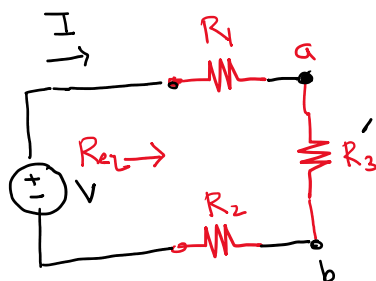
$$a-c: R_4'$$

$$R_5' = R_4' \parallel R_5$$

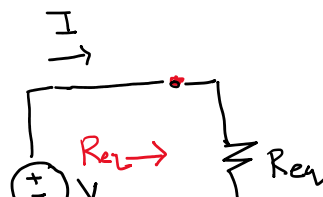
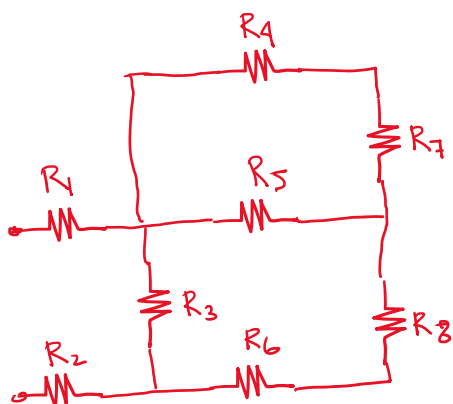


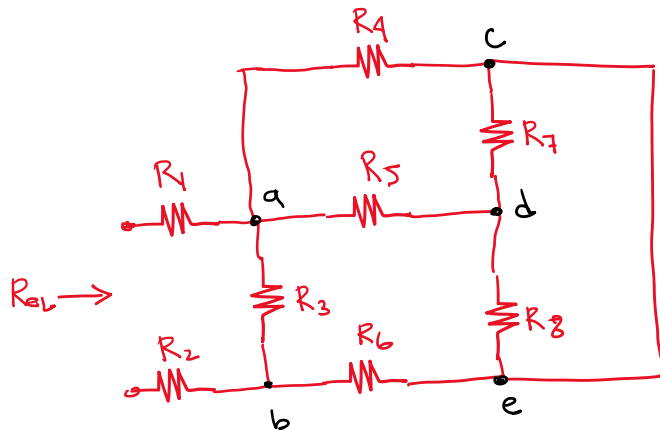
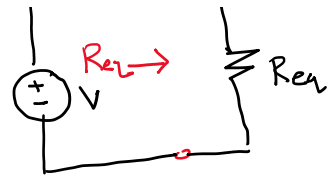
$$a-b: R_3 \text{ and } R_5' + R_6' = R_9$$

$$a-b: R_3' = R_3 \parallel R_9$$

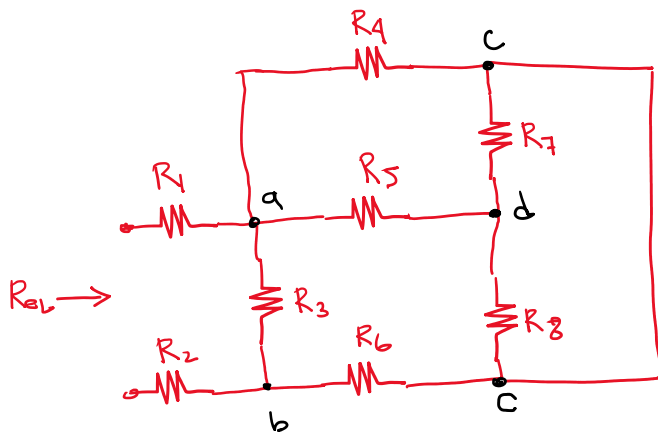


$$R_{eq} = R_1 + R_2 + R_3'$$



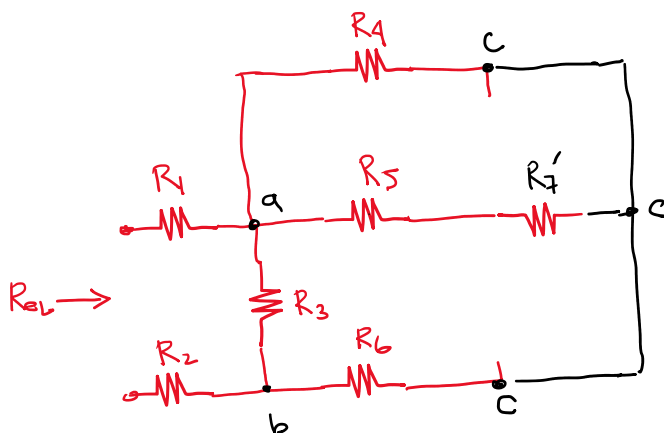


betⁿ c & e: No element, they are
Same point same node



c & d: R_7 , c & e: R_8

$$R_7' = R_7 \parallel R_8$$

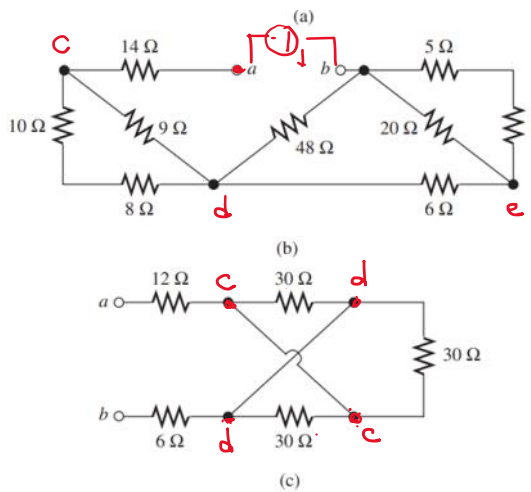
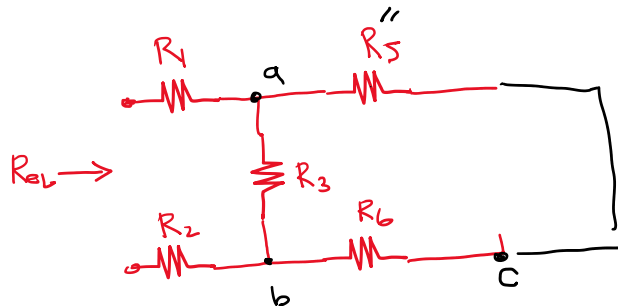


a.c. $R_5' = R_5 + R_7'$

a-c: R_R

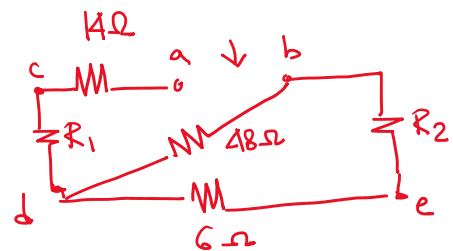
a-c $R_5'' = R_4 \parallel R_5'$

$$R_{eq} = R_2 + R_1 + [R_3 \parallel (R_5'' + R_6)]$$



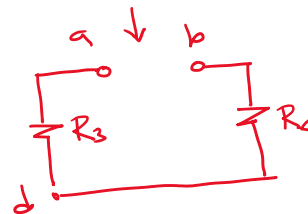
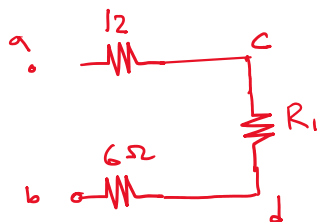
$$R_1 = (10 + 8) \parallel 9$$

$$R_2 = (5 + 15) \parallel 20$$

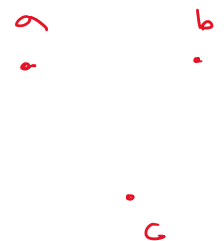
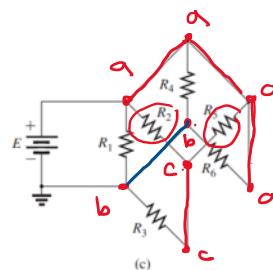
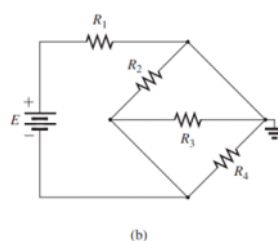
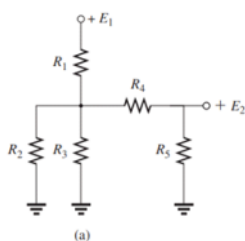
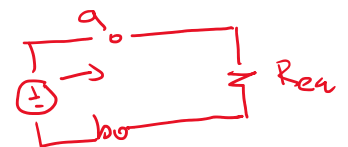


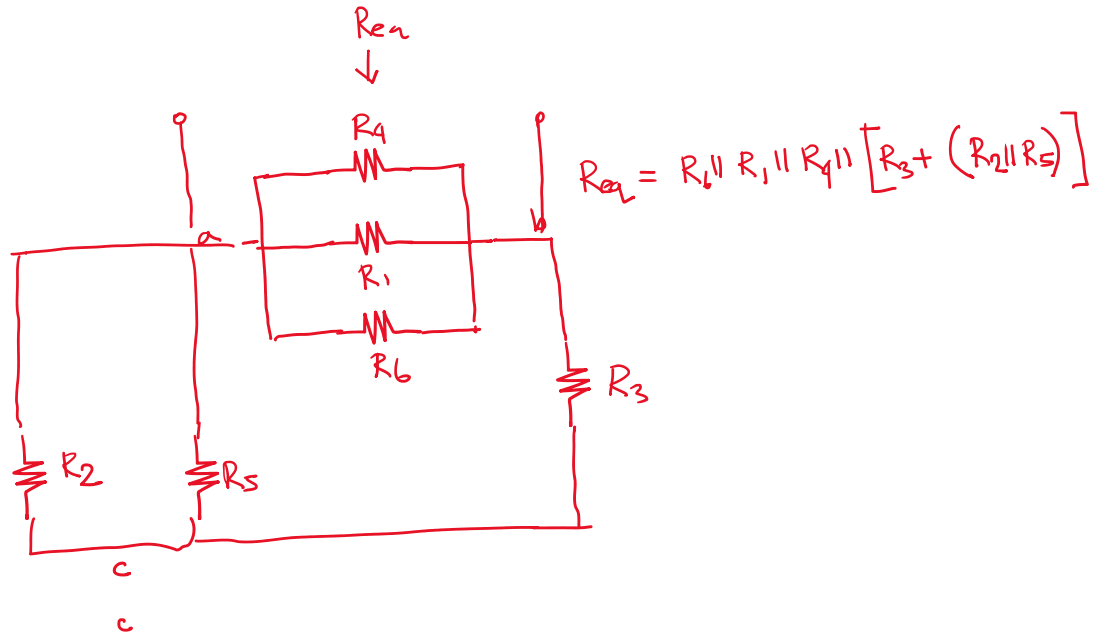
$$R_3 = R_1 + 14, \quad b-d: 48 \parallel (R_2 + 6) = R_4$$

$$c-d: 30 \parallel 30 \parallel 30 = R_1$$



$$R_{eq} = R_3 + R_4$$





Calculate the equivalent resistance R_{ab} in the circuit in Fig. 2.37.

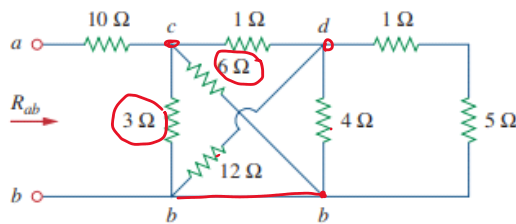
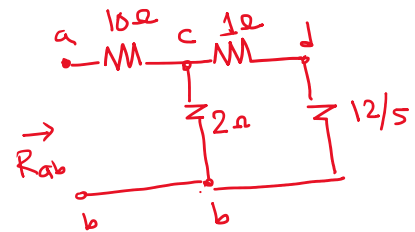


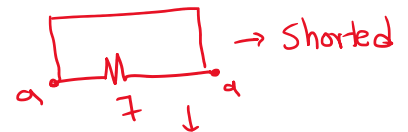
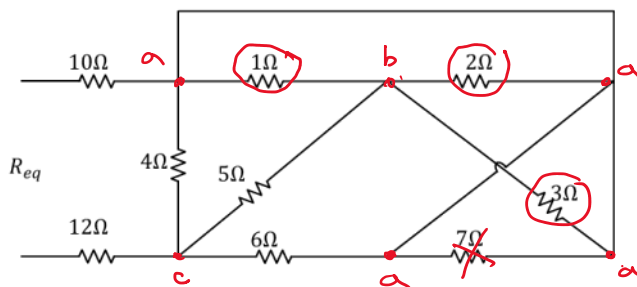
Figure 2.37
For Example 2.10.

$b-d: 4 \parallel 6 \parallel 12 = 12/5 \Omega$

$b-c: 6 \Omega \parallel 3 \Omega = 2 \Omega$

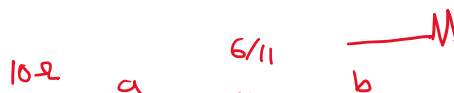


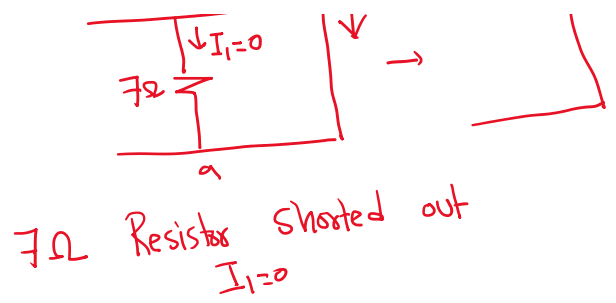
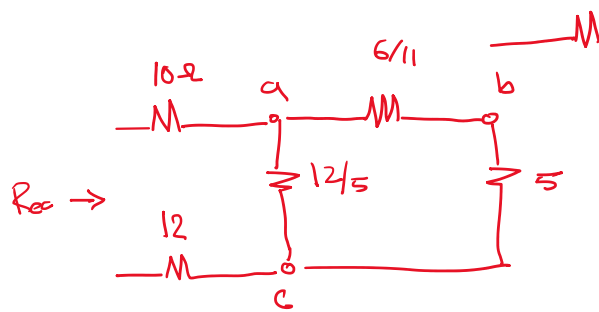
$R_{eq} = 10 \Omega + [2 \parallel (1 + 12/5)]$



$a-b: 1 \parallel 2 \parallel 3 = 6/11$

$a-c = 4 \parallel 6 = 12/5$





$$R_{eq} = \left[(6/11 + 5) \parallel 12/5 \right] + 10 + 12$$